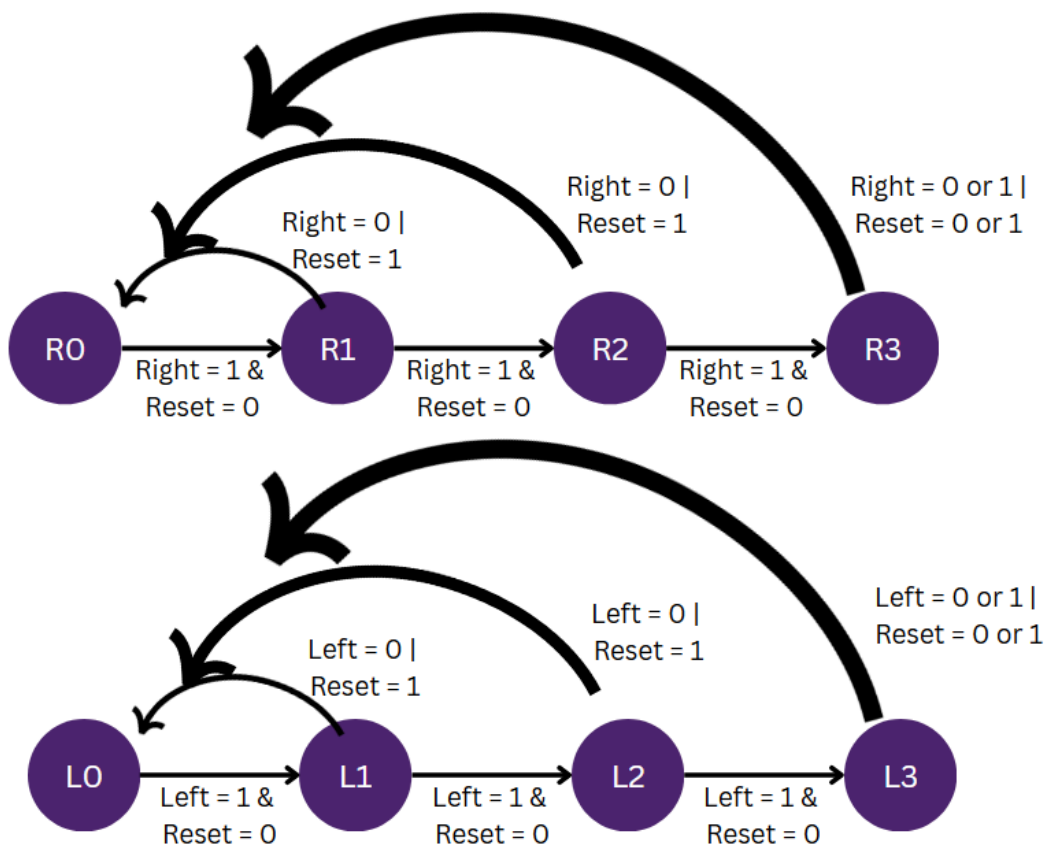


■ (1 pt) Please indicate how many hours you spent on this lab. This will be helpful for calibrating the workload for next time the course is taught. Include any suggestions you have for improving the lab. You will get one point for listing your time on each lab.

Lab 1: 2 Hours
Lab 2: 2 Hours
Lab 3: 2 Hours
Lab 4: 3 Hours
Lab 5: 3 Hours
Lab 6: 4 hours
Lab 7: 2 Hours
Lab 8: 2 Hours
Lab 9: 1 Hours
Lab 10: 4 Hours

2. (14 pts total) Thunderbird FSM

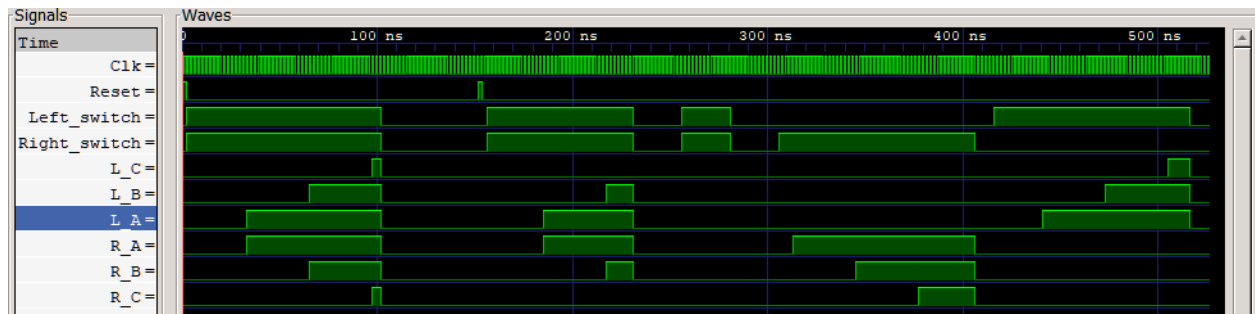
■ (3 pts) Submit a clear sketch of your Thunderbird FSM state transition diagram.



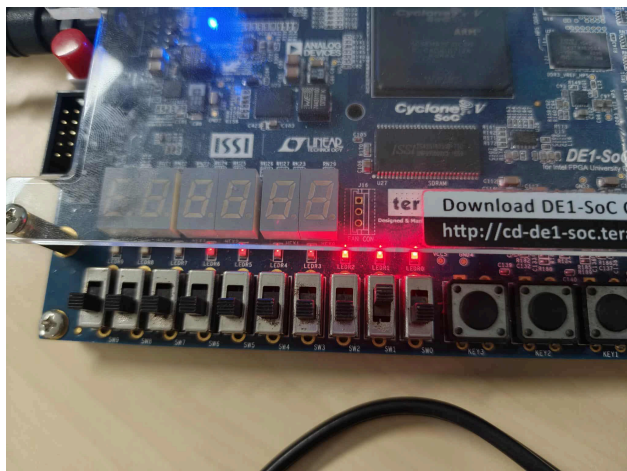
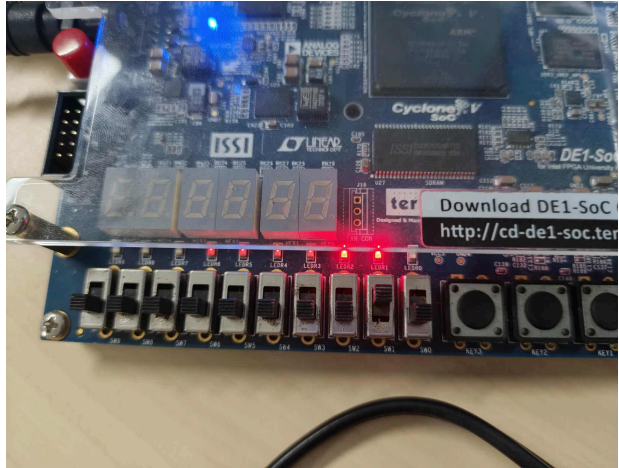
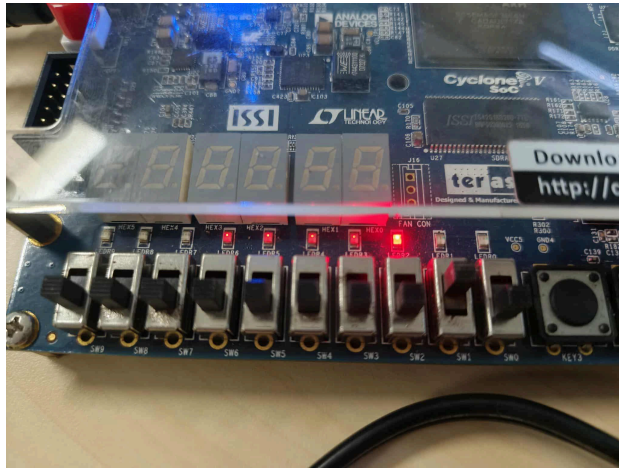
■ (2 pts) Submit thunderbird.v (as a separate file from your PDF submission).

- (2 pts) Submit a screenshot of your Thunderbird FSM state transition diagram timing waveform.

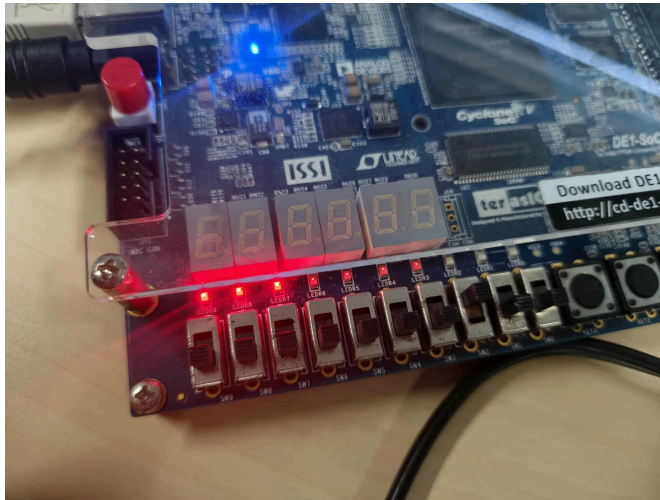
Be sure to display the signals in this order (from top to bottom): clk, reset, left, right, LC, LB, LA, RA, RB, RC.



- (2 pts) Demonstrate your FSM displaying RA, then RA and RG, then RA, RB, and RB on the right-most LEDs (3 pictures). Clearly also show the “right” switch (SW[1]) being asserted.



- (1 pt) Demonstrate your FSM displaying LA, LB, and LC on LEDR[5:3] (1 picture). Clearly also show the “left” switch (SW[2]) being asserted.



- (2 pts) Briefly describe how you tested the system on the DE1-SoC board and whether it worked according to the specifications. Did you observe switch bounce?
I tested my code by following the inputs that I assigned in my test bench, it worked as expected and showed no error in normal operation. Switch bounce occurred when I switched either left or right from off to on or enabling and disabling the reset switch. The lights will reset back to their initial state of all off even if it was not the next sequence.
- (2 pts) Write a brief paragraph about what you learned in this lab.
I learned how to make a FSM and how it can be applied in real life situations. The Thunderbird was a really good way to explaining the use of counters and is a good model for creating a FSM/ The use of flip flops and clock dividers help me understand how to make physical counters without relying on premade hardware.
- Please indicate any bugs you found in these lab instructions, or any suggestions you have to improve the lab.